Stochastic optimization

Exercise collection

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This collection accompanies the lecture slides. Each exercise names the deck it depends on. This version holds the *statements only*; the solutions are in `exercises_solutions.pdf`.

Sources are indicated exercise by exercise; several are adapted from Birge and Louveaux, *Introduction to Stochastic Programming*, 2nd edition, Springer, 2011.

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1 Generalities

Exercise 1.1. *(Deck 03.) Show that the value of the stochastic solution and the expected value of perfect information are always non-negative.*

2 Two-stage problems with recourse

Exercise 2.1. *(Deck 01 and 03.) Consider the farmer problem of deck 01. Is its recourse* complete*?* Relatively complete*?* Simple*?*

Exercise 2.2. *(Deck 01.) A newsvendor buys each copy at cost $c = 2$, sells it at price $q = 5$, and gets $r = 1$ back for each unsold copy. The daily demand is $\boldsymbol{\xi} \sim U[0, 100]$.*

(a) Having already bought t copies, what is the expected marginal revenue of one more copy? Deduce the optimal order quantity x^*.

(b) Compare x^* with the expected demand, and explain the gap in terms of the cost of a missed sale and of an unsold copy.

Exercise 2.3. *(Deck 03; Birge and Louveaux, p. 87.) Consider the second-stage program*

$$Q(x, \xi) = \min_y \{y \mid \xi y = 1 - x,\ y \geq 0\},$$

where $\boldsymbol{\xi}$ follows a triangular distribution on $[0, 1]$, i.e. $P[\boldsymbol{\xi} \leq u] = u^2$.

(a) Is the recourse fixed? Why?

(b) Express $K_2(\xi)$ for every $\xi \in [0, 1]$.

(c) Express K_2, K_2^P and $\mathcal{Q}$.

Exercise 2.4. *(Deck 03.) Consider the second-stage program*

$$\begin{aligned} \min_y\ & 2y_1 + y_2 \\ \text{s.t. }\ & y_1 - y_2 \leq 2 - \boldsymbol{\xi} x_1, \\ & y_2 \leq x_2, \\ & y_1, y_2 \geq 0. \end{aligned}$$

Find $K_2(\xi)$ and K_2 when (a) $\boldsymbol{\xi} \sim U[0, 1]$; (b) $\boldsymbol{\xi} \sim$ Poisson(λ), $\lambda > 0$. Which properties do you expect of K_2? Is the recourse simple or complete? If not, under which conditions would it be relatively complete?

Exercise 2.5. *(Deck 03; Birge and Louveaux, p. 102.) Consider the second-stage program*

$$\begin{aligned} \min_y\ & 2y_1 + y_2 \\ \text{s.t. }\ & y_1 + y_2 \geq 1 - x_1, \\ & y_1 \geq \boldsymbol{\xi} - x_1 - x_2, \\ & y_1, y_2 \geq 0. \end{aligned}$$

Show that this program has a complete recourse when $\boldsymbol{\xi}$ has a finite expectation.

Exercise 2.6. *(Deck 03.) A company must order a quantity x of a product to meet a demand d that is unknown at the time of ordering. The unit ordering cost is $c > 0$. If $d > x$, a penalty $b \geq 0$ is incurred per unit of unmet demand; if $d < x$, a holding cost h is incurred per surplus unit.*

(a) Formulate the problem as a two-stage stochastic program.

(b) Is the recourse simple, complete, relatively complete?

(c) What can be expected of the first-stage solution when $b < c$?

Exercise 2.7. *(Deck 03.) A good X must be produced before its demand is known. Each unit produced costs \$2. Demand is discrete, taking the values D_s with probabilities p_s, $s = 1, \ldots, S$. Demand can also be met by buying from an external supplier at \$3 per unit, once it has been observed. How much should*

be produced now? Formulate the problem as a two-stage stochastic program minimizing the expected production cost.

Exercise 2.8. *(Deck 03; Higle,* Tutorials in Operations Research*, INFORMS, 2005.) The Dakota Furniture Company makes desks, tables and chairs. Each item consumes lumber and two kinds of skilled labour, finishing and carpentry:*

Resource	*Cost*	*Desk*	*Table*	*Chair*
Lumber (board ft.)	*$2*	*8*	*6*	*1*
Finishing (hours)	*$4*	*4*	*2*	*1.5*
Carpentry (hours)	*$5.2*	*2*	*1.5*	*0.5*
Selling price		*$60*	*$40*	*$10*

Resources must be acquired before the demand is known. Three equally structured demand scenarios are considered, with probabilities 0.3*,* 0.4 *and* 0.3*:*

Scenario	*Desks*	*Tables*	*Chairs*
Low ($p = 0.3$)	*50*	*20*	*200*
Most likely ($p = 0.4$)	*150*	*110*	*225*
High ($p = 0.3$)	*250*	*250*	*500*

Formulate the problem as a two-stage stochastic program maximizing the expected profit. Only the formulation is asked.

Exercise 2.9. *(Deck 03; Birge and Louveaux, p. 142.) Consider*

$$\begin{aligned} \min_x \; & x_1 + 4x_2 + \mathbb{E}_{\boldsymbol{\xi}}[Q(x, \boldsymbol{\xi})] \\ \text{s.t. } & x_1 + x_2 = 1, \quad x \geq 0, \end{aligned}$$

with

$$\begin{aligned} Q(x, \xi) = \min_y \; & y_1 + 10y_2 + 10y_3 \\ \text{s.t. } & y_1 + y_2 - y_3 = \xi + x_1 - 2x_2, \\ & y_1 \leq 2, \qquad y \geq 0, \end{aligned}$$

and $\boldsymbol{\xi} \sim U[1, 3]$.

(a) Is the recourse fixed, simple, complete, relatively complete? Answer true or false for each, with a brief justification.

(b) For given x *and* ξ*, show, by discussing the value of* $\xi + x_1 - 2x_2$*, that*

$$y^*(x, \xi) = \begin{cases} y_1 = \xi + x_1 - 2x_2, \; y_2 = y_3 = 0 & \text{if } 0 \leq \xi + x_1 - 2x_2 \leq 2, \\ y_1 = 2, \; y_2 = \xi + x_1 - 2x_2 - 2, \; y_3 = 0 & \text{if } \xi + x_1 - 2x_2 > 2, \\ y_3 = 2x_2 - \xi - x_1, \; y_1 = y_2 = 0 & \text{if } \xi + x_1 - 2x_2 < 0. \end{cases}$$

(c) Which expected property of $Q(x, \xi)$ *does* $y^*(x, \xi)$ *confirm?*

(d) Is $\mathcal{Q}(x) = \mathbb{E}_{\boldsymbol{\xi}}[Q(x, \boldsymbol{\xi})]$ *differentiable? Why?*

(e) Express K_1*,* $K_2(\xi)$*,* K_2^P *and* K_2*.*

(f) Give the deterministic equivalent (extensive form) of the discretized problem where $\boldsymbol{\xi}$ *takes the values* 1*,* 2*,* 3*.*

3 Decomposition methods

Exercise 3.1. *(Deck 04 and LP background.) Apply the dual decomposition method (the L-shaped method with a single scenario) to*

$$\min_{0\le x\le 5}\ 2x + f(x), \qquad f(x) = \min_{y\ge 0}\{y_1 + 3y_2 \mid y_1 + y_2 = 4 - x,\ \ y_1 + y_3 = 3\}\,.$$

A demand $4-x$ must be met exactly by a cheap source y_1 of capacity 3 (y_3 is the slack) and by an expensive source y_2. The master problem is $\min\{2x+\theta \mid 0\le x\le 5,\ \theta\ge 0,\ \text{cuts}\}$, the constraint $\theta\ge 0$ being valid since $f\ge 0$.

(a) Identify W, q, h and T, and write the dual of the second-stage problem at $\hat{x}$.

(b) Suppose the master returns $\hat{x}=5$. Using Farkas' lemma, show that the second stage is infeasible by exhibiting a certificate σ, and write the feasibility cut in terms of x.

(c) At the first iteration, $(\hat{x},\hat{\theta})=(0,0)$. Compute $f(0)$ and an optimal dual solution $\hat{\pi}$ (hint: complementary slackness), write the optimality cut, and decide whether the algorithm stops.

(d) Why is the master's value $2\hat{x}+\hat{\theta}$ a lower bound on the optimal value at every iteration?

(e) Carry the method to its end: give the successive iterates, cuts and bounds, and the optimal solution.

Exercise 3.2. *(Deck 04.) Consider*

$$\begin{aligned}
\min\ & 100x_1 + 150x_2 + \mathbb{E}_{\boldsymbol{\xi}}[q_1y_1 + q_2y_2]\\
\text{s.t. }\ & x_1 + x_2 \le 120,\\
& 6y_1 + 10y_2 \le 60x_1, \qquad 8y_1 + 5y_2 \le 80x_2,\\
& y_1 \le d_1, \qquad y_2 \le d_2,\\
& x_1 \ge 40,\ x_2 \ge 20, \qquad y_1, y_2 \ge 0,
\end{aligned}$$

under two scenarios $\boldsymbol{\xi} = (d_1, d_2, q_1, q_2)$: $\xi_1 = (500, 100, -24, -28)$ with probability $p_1 = 0.4$, and $\xi_2 = (300, 300, -28, -32)$ with probability $p_2 = 0.6$.

Run the first iteration of the single-cut L-shaped method, showing that the second-stage solutions are respectively $y^* = (137.5, 100)$ and $y^* = (80, 192)$. To obtain the dual solutions, use complementary slackness: a dual variable can be non-zero only if the associated primal constraint is active.

Exercise 3.3. *(Deck 04; Birge and Louveaux, p. 174.) Consider the problem*

$$\begin{aligned}
\min_{x\ge 0}\ & 3x_1 + 2x_2 + \mathbb{E}_{\boldsymbol{\xi}}\left[\min_{y\ge 0}\ -15y_1 - 12y_2\right]\\
\text{s.t. }\ & 3y_1 + 2y_2 \le x_1, \qquad 2y_1 + 5y_2 \le x_2,\\
& 0.8\,\xi_1 \le y_1 \le \xi_1, \qquad 0.8\,\xi_2 \le y_2 \le \xi_2,
\end{aligned}$$

where $\boldsymbol{\xi}$ takes the values $\xi^{(1)} = (4, 6)$, $\xi^{(2)} = (4, 8)$, $\xi^{(3)} = (6, 6)$ and $\xi^{(4)} = (6, 8)$, each with probability $1/4$. Started from $x^1 = (0, 0)$, the L-shaped method needs three feasibility cuts, $3x_1 + x_2 \ge 123.2$, then $x_2 \ge 22.4$ and $x_2 \ge 41.6$, to reach a point of K_2, $x^4 = (27.2, 41.6)$.

(a) Show that every $x \in K_2$ satisfies $x_1 \ge 27.2$ and $x_2 \ge 41.6$.

(b) These two constraints are added to the first stage from the start. Can this exclude the optimal solution? Show that the recourse becomes relatively complete.

(c) Is the recourse then complete?

(d) The method, run without these two constraints, never generates $x_1 \geq 27.2$. Why does it still reach a point of K_2, and why does it need three cuts where two constraints suffice?

Exercise 3.4. *(Deck 04.) True or false? Justify your answer.*

(a) If two solvers, applied to the same phase-1 problem at the same point x^ν, return two different feasibility cuts, one of them is wrong.

(b) Like a feasibility cut, an optimality cut removes first-stage decisions x from the master problem.

Exercise 3.5. *(Deck 04.) In the regularized decomposition method of Ruszczyński, what is the difference between an* exact *serious step and an* approximate *serious step? Why must the two be distinguished?*

4 Multistage problems

Exercise 4.1. *(Decks 06 and 08; Heitsch, Römisch and Strugarek,* Stability of multistage stochastic programs*, SIAM Journal on Optimization 17(2):511–525, 2006.) Consider the stylized multistage program below, seeking an optimal purchasing strategy under uncertain costs $\boldsymbol{\xi}_t$ over the stages $t = 1, \ldots, H$. The amounts purchased are x_t, and a prescribed amount a must be reached at the horizon H:*

$$\begin{aligned} &\min_x \mathbb{E}\left[\sum_{t=1}^{H} \boldsymbol{\xi}_t x_t\right], \\ &\text{s.t. } s_t - s_{t-1} = x_t, \quad t = 1, \ldots, H, \\ &\qquad s_0 = 0, \quad s_H = a, \quad x_t \geq 0, \quad s_t \geq 0, \end{aligned}$$

where s_t is a state variable holding the amount accumulated at the end of stage t.

Take three stages ($t = 1, 2, 3$), a binary tree, and $\xi_1 = 10$; the cost goes up by 1 with probability 0.6 and down by 1 with probability 0.4.

(a) Draw the scenario tree.

(b) Write the problem (i) in extensive form, (ii) using a scenario decomposition.

(c) In which of the two formulations must nonanticipativity constraints be written explicitly?

Exercise 4.2. *(Deck 06.) A firm produces the goods $j \in N$ over the periods $t \in \mathcal{T} = \{1, \ldots, H\}$, with the resources $i \in M$, among which the labor $L \in M$. One unit of j requires a_{ij} units of resource i, of which b_{it} are available in period t; up to U_{it} more can be bought, at the unit cost β_{it}. Producing one unit of j in period t costs α_{jt}, and changing the workforce, identified with the labor required by the production, costs γ_t^+ per unit of increase and γ_t^- per unit of decrease from period $t-1$ to period t. The demand ω_{jt}*

of each period is random and must be met: a missing unit is bought from an external source at the cost δ^-_{jt}, an unsold unit is stored at the cost δ^+_{jt}. The model is

$$\begin{aligned}
\min\ & \sum_{j,t} \alpha_{jt} x_{jt} + \sum_{i,t} \beta_{it} u_{it} + \sum_{t=2}^{H} \left(\gamma_t^+ z_t^+ + \gamma_t^- z_t^-\right) + \mathbb{E}_{\boldsymbol{\omega}}\Big[\min \sum_{j,t} \left(\delta^+_{jt} y^+_{jt} + \delta^-_{jt} y^-_{jt}\right)\Big] \\
\text{s.t.}\ & \sum_{j\in N} a_{ij} x_{jt} \le b_{it} + u_{it}, \quad 0 \le u_{it} \le U_{it}, \quad i \in M,\ t \in \mathcal{T}, \\
& z_t^+ - z_t^- = \sum_{j\in N} a_{Lj}(x_{jt} - x_{j,t-1}), \quad t = 2, \dots, H, \qquad x, z^{\pm} \ge 0, \\
& x_{jt} + y^+_{j,t-1} - y^+_{jt} + y^-_{jt} = \omega_{jt}, \quad j \in N,\ t \in \mathcal{T}, \qquad y^{\pm} \ge 0,
\end{aligned}$$

where y^+_{jt} is the stock of product j at the end of period t (the initial stock y^+_{j0} is given) and y^-_{jt} the amount bought from the external source in period t. All the costs are nonnegative.

(a) Which decisions are made here and now, and which are recourse decisions? What information may each recourse decision use?

(b) Suppose that buying ahead never pays: $\delta^-_{jt} + \delta^+_{jt} \ge \delta^-_{j,t+1}$ for every j and $t < H$. Show that, for a fixed production plan, buying only the deficit,

$$y^-_{jt} = \max\{0,\ \omega_{jt} - x_{jt} - y^+_{j,t-1}\}, \qquad y^+_{jt} = \max\{0,\ y^+_{j,t-1} + x_{jt} - \omega_{jt}\},$$

is optimal in every scenario. What does this imply for the nonanticipativity constraints of the extensive form?

(c) Take one product, two periods, no production ($b = U = 0$, hence $x = 0$), no initial stock, $\omega_1 = 0$, and $\omega_2 = 0$ or 10 with probability $1/2$; the purchase costs are $\delta^-_1 = 1$ and $\delta^-_2 = 4$, the storage costs $\delta^+_1 = \delta^+_2 = 0.5$. Compute the optimal purchase in period 1 with nonanticipativity and with perfect information, and the cost of the policy of (b). Conclude.

(d) The balance of period t involves x_{jt}, decided at the first stage, and $y^+_{j,t-1}$, decided at the previous stage. Write the problem in the form of the multistage linear programs of deck 06, written here $W_t X_t = h_t(\xi_t) - T_{t-1}(\xi_t) X_{t-1}$ with X_t the decisions of stage t (x being the production), in which each stage depends on the previous one only. Where does the randomness enter?

(e) (Optional, Julia.) Write a builder for `code/nested_benders.jl`, and solve a small instance on a scenario tree; compare with the extensive form.

5 Random number generation

Exercise 5.1. *(Deck 02.) A linear congruential generator produces $x_i = (a x_{i-1} + c) \bmod m$.*

(a) For $m = 16$ and $c = 1$, use the period theorem to find every a giving a full period.

(b) The "standard minimal" generator of Park and Miller is $x_i = 16807\, x_{i-1} \bmod (2^{31} - 1)$. Why can its period not be m? What is the largest it could be, and why is that value actually reached here?

(c) *For $m = 16$, which of the pairs $(a, c) = (3, 4)$, $(5, 3)$, $(7, 3)$, $(4, 1)$ give a full period? For $(4, 1)$, write the sequence from $x_0 = 0$: what happens, and why?*

Exercise 5.2. *(Deck 02.) Consider the linear feedback shift register over $\mathbb{F}_2$ with $k = 4$ and $x_n = x_{n-1} \oplus x_{n-4}$, seeded at 0001.*

(a) *Why is the all-zero state excluded, and why can the period not exceed $2^k - 1$?*

(b) *A Tausworthe generator reads the output stream w bits at a time, spaced ν apart. The slides take $w = 3$, $\nu = 3$ and observe only five distinct values instead of fifteen. Explain, and find every $\nu \in \{1, \dots, 14\}$ giving the maximal output period.*

Exercise 5.3. *(Deck 02.) Let X take the values 1, 2, 3 with probabilities 0.2, 0.5, 0.3.*

(a) *Write F_X and its generalized inverse, and check that $F_X^{-1}(U)$, $U \sim U[0, 1]$, has the right distribution.*

(b) *Why can the ordinary inverse not be used here?*

(c) *Inversion is called the "gold standard" in the slides. Which property makes common random numbers and antithetic variates reliable with it?*

(d) *A demand Y is zero with probability $1/2$ and, otherwise, uniform on $[0, 2]$. Write F_Y and its generalized inverse, and give the values generated for $U = 1/4$ and $U = 3/4$. Which values of U produce $Y = 0$, and why is that correct?*

Exercise 5.4. *(Deck 02.) In acceptance–rejection one takes $t \geq f$ with $\int t = a$, so that $r = t/a$ is a density, then repeats: draw $Y \sim r$ and $U \sim U[0, 1]$ until $U \leq f(Y)/t(Y)$, and return Y.*

(a) *Compute the probability that one iteration is accepted.*

(b) *Deduce the expected number of iterations, and explain why the slides ask for a close to 1.*

(c) *Why does this method break common random numbers, even when the accepted value is correct?*

Exercise 5.5. *(Deck 02.) Let C be a bivariate copula, so that $H(x, y) = C(F_X(x), F_Y(y))$ is the joint cdf of (X, Y).*

(a) *Show that the independence copula $C(u, v) = uv$ satisfies the Fréchet–Hoeffding bounds.*

(b) *The upper bound is attained by $C(u, v) = \min(u, v)$. Which dependence between X and Y does it describe?*

(c) *A tempting proof of the upper bound argues $P[A \cap B] \leq P[A]\,P[B] \leq \min(P[A], P[B])$. Give a counterexample to the first inequality, and state the correct argument.*

6 Monte Carlo approximation

Exercise 6.1. *(Decks 03 and 09.) Show that*

$$\mathbb{E}_{\boldsymbol{\xi}}\left[\min_{x \in \mathcal{X}} z(x, \boldsymbol{\xi})\right] \leq \min_{x \in \mathcal{X}} \mathbb{E}_{\boldsymbol{\xi}}\left[z(x, \boldsymbol{\xi})\right]$$

for a deterministic feasible set $\mathcal{X}$. How is this result to be interpreted for stochastic programming? Extend it to a Monte Carlo sample average.